Day 122 | Unit 09 | Teacher plan

Display, Describe, and Summarize Data

Summarize Data with Mean and Median

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c

Learning target: I can summarize Data with Mean and Median, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. The mean is a fair share: add all values and divide by the number of observations. Report the count and measurement units. A large unusual value can pull the mean toward it. Order the data first. For an odd count, choose the middle value. For an even count, average the two middle values. The median often resists an unusually high or low value better than the mean.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references:

Day 122 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Summarize Data with Mean and Median

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c

Learning target: I can summarize Data with Mean and Median, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

The mean is a fair share: add all values and divide by the number of observations. Report the count and measurement units. A large unusual value can pull the mean toward it. Order the data first. For an odd count, choose the middle value. For an even count, average the two middle values. The median often resists an unusually high or low value better than the mean.

Worked example 1

Reading times in minutes are 5, 6, 6, 7, 7, 8. Report n, units, and mean.

1. The mean is a fair share: add all values and divide by the number of observations.

2. Report the count and measurement units.

3. Sum = 39

4. mean = 39 ÷ 6 = 13/2.

5. State the result in the form requested: n = 6; minutes; mean 13/2 min.

Answer: n = 6; minutes; mean 13/2 min

Worked example 2

Six students recorded reading times (minutes): 4, 6, 6, 8, 8, 10. Select all true statements.
A. The median is always the largest observation.
B. The median is 7 minutes.
C. The median is 8 minutes.
D. The median is the average of the 3rd and 4th ordered values.

1. Order the data first.

2. Six ordered observations have middle positions 3 and 4, so median = (6 + 8)/2 = 7.

Answer: B, D - The median is 7 minutes.; The median is the average of the 3rd and 4th ordered values.

Day 122 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Report n, units, and mean.

02 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Find the median. Show the middle positions.

03 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Report n, units, and mean.

04 / Constructed response

Reading times in minutes are 8, 12, 12, 16, 16, 20. Find the median. Show the middle positions.

05 / Constructed response

Reading times in minutes are 5, 8, 8, 11, 11, 14. Report n, units, and mean.

06 / Constructed response

Reading times in minutes are 3, 6, 6, 9, 9, 12. Find the median. Show the middle positions.

07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Report n, units, and mean. If one more student reads 6 minutes, find the new mean.

08 / Multiple select

Six students recorded reading times (minutes): 3, 8, 8, 13, 13, 18. Select all true statements.
A. The median is the average of the 3rd and 4th ordered values.
B. The median is 23/2 minutes.
C. The median is always the largest observation.
D. The median is 21/2 minutes.

09 / Two-part numeric entry

Six students recorded reading times (minutes): 3, 9, 9, 15, 15, 21. Part A: enter the mean. Part B: a seventh student reads 27 minutes. Enter the new mean and explain the denominator.

10 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 10, 19. Find the median and mean. Explain the difference in this set.

Day 122 | Unit 09 | Answer key

Display, Describe, and Summarize Data

Summarize Data with Mean and Median

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c

Learning target: I can summarize Data with Mean and Median, and explain and check my reasoning.

1. n = 6; minutes; mean 11/2 min

Sum = 33; mean = 33 ÷ 6 = 11/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

2. Median 11/2 min

6 ordered values: average positions 3 and 4: (5 + 6) ÷ 2 = 11/2.

6.SP.B.5c | Constructed response

3. n = 6; minutes; mean 9/2 min

Sum = 27; mean = 27 ÷ 6 = 9/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

4. Median 14 min

6 ordered values: average positions 3 and 4: (12 + 16) ÷ 2 = 14.

6.SP.B.5c | Constructed response

5. n = 6; minutes; mean 19/2 min

Sum = 57; mean = 57 ÷ 6 = 19/2.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

6. Median 15/2 min

6 ordered values: average positions 3 and 4: (6 + 9) ÷ 2 = 15/2.

6.SP.B.5c | Constructed response

7. n = 6; minutes; mean 31/6 min; new mean 37/7 min

Sum = 31; mean = 31 ÷ 6 = 31/6. New sum = 37; divide by 7.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Constructed response

8. A, D - The median is the average of the 3rd and 4th ordered values.; The median is 21/2 minutes.

Six ordered observations have middle positions 3 and 4, so median = (8 + 13)/2 = 21/2.

6.SP.B.5c | Multiple select

9. 12 min; new mean 99/7 min

Original sum 72 divided by 6 gives 12. Add 27, then divide the new sum by 7 because there are now 7 observations.

6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c | Two-part numeric entry

10. Median 17/2 min; mean 61/6 min; the high value pulls the mean upward

6 ordered values: average positions 3 and 4: (8 + 9) ÷ 2 = 17/2.

6.SP.B.5c | Constructed response